(i) n, the number of trials is infinitely large, i.e. $n \to \infty$

(ii) neither p (nor q) is very small,

The normal distribution of a variable when represented graphically, takes the shape of a symmetrical curve, known as the Normal Curve. The curve is asymptotic to x-axis on its either side.

Chief Characterisitics or Properties of Normal Probabilty distribution and Normal probability Curve .

The normal probability curve with mean μ and standard deviation σ has the following properties :

(i) the curve is bell- shaped and symmetrical about the line $x=\mu$

(ii) Mean, median and mode of the distribution coincide.

(iii) x – axis is an asymptote to the curve. (tails of the cuve never touches the horizontal (x) axis)

(iv) No portion of the curve lies below the x-axis as $f(x)$ being the probability function can never be negative.

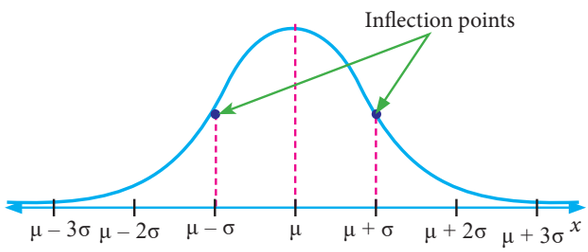


Fig 7.2

(v) The Points of inflexion of the curve are $x = \mu \pm \sigma$

(vi) The curve of a normal distribution has a single peak i.e it is a unimodal.

(vii) As x increases numerically, $f(x)$ decreases rapidly, the maximum probability occurring at the point x = μ and is given by $[p(x)]\max = 1/\sigma\sqrt{2\pi}$

(viii) The total area under the normal curve is equal to unity and the percentage distribution of area under the normal curve is given below

(a) About 68.27% of the area falls between μ – σ and μ + σ

$P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

(b) About 95.5% of the area falls between μ – 2σ and μ + 2σ

$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$

(c) About 99.7% of the area falls between μ – 3σ and μ + 3σ

$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$

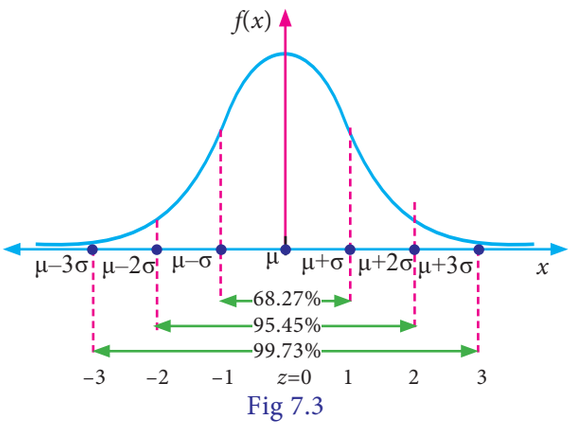


Fig 7.3

Standard Normal Distribution

A random variable Z = (X–μ)/σ follows the standard normal distribution. Z is called the standard normal variate with mean 0 and standard deviation 1 i.e Z ~ N(0,1). Its Probability density function is given by :

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad -\infty < z < \infty$$

1. The area under the standard normal curve is equal to 1.

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